
Biology in silico: The Bioinformatics Revolution

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Biology in *silico*:

THE BIOINFORMATICS REVOLUTION

This is not the end.

It is not even the beginning of the end.

But it is, perhaps, the end of the beginning.

—SIR WINSTON CHURCHILL

MARK BLOOM

This quotation refers to the Second World War, but also can be applied to the effort to decode and understand the human genome. The Human Genome Project (HGP) was initiated by the Department of Energy (DOE) in 1986 and the National Institutes of Health (NIH) joined the project two years later. Genome projects also were begun in a number of other industrialized countries, notably, the United Kingdom. On June 26, 2000 Dr. Francis Collins, director of the National Human Genome Research Institute and Dr. Craig Venter, president of Celera Genomics, issued a joint announcement saying that they had independently obtained rough drafts of the human genome (Wade 2000). The finished sequence is scheduled for completion sometime in 2003. This will mark the official end of the effort to decode the entire set of human genes. It also will serve to mark the unofficial start of the next phase of our continuing quest to understand how genetics contributes to our health and well-being.

The HGP is estimated to cost approximately \$3 billion over a 15-year period. From the outset, the project was controversial, both within and outside the scientific community. Many of the scientists' reservations had to do with money. They were concerned that funds spent on the HGP would take away from monies earmarked for basic research. Other objections, however, were of a more scientific nature.

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To Sequence or Not To Sequence?

Critics noted that the portion of the human genome that actually codes for genes was thought to be less than 5 percent of the total. The remainder was dismissed as "junk" and not worth the cost of sequencing. Still, other scientists like Sydney Brenner pointed out that "We have the surprising result that most of the human genome is junk; junk and not garbage because there is a difference that everybody knows; junk is kept while garbage is thrown away" (Brenner 1990). Nobel laureate Paul Berg is another scientist who believes that junk DNA has many lessons to teach us. He once gave a talk at the Salk Institute where he described how the SV40 virus can produce two different RNA transcripts from the same DNA sequence. He predicted that this strategy of "alternate splicing" also would have application in mammalian genomes. Francis Crick disagreed, and the two made a bet of two cases of wine. Berg made similar bets with other colleagues and now has a well-stocked wine cellar (Davis 1990).

As the HGP moved forward, other scientists worried that the resultant torrent of sequence data would prove overwhelming and unintelligible. Robert Weinberg, who at the time was studying genes associated with retinoblastoma, lamented, "I fear . . . that the important discussions have already been made and that the great sequencing juggernaut will soon begin its inexorable forward motion, flooding our desks with oceans of data whose scope defies conception and our ability to interpret meaningfully" (Lee 1991). During the late 1980s when

Weinberg made this remark, the ability of computers to analyze sequence data and identify genes or establish gene functions was quite limited. The last decade has seen explosive growth in the young science of bioinformatics, and we no longer hear the complaint that “we don’t know what to do with all this sequence data.”

The Birth of Bioinformatics

The explosion of data produced by the Human Genome Project has called forth the creation of a new discipline – bioinformatics, whose focus is on the acquisition, storage, analysis, modeling, and distribution of the many types of information embedded in DNA and protein sequence data.

Lee Rowen, Gregory Mahairas, and Leroy Hood, *Science*, October 24, 1997

The evolution of ideas in biology can be likened to the punctuated equilibria model put forward by Niles Eldredge and Stephen J. Gould (Eldredge & Gould 1972). According to this view, species (or in my analogy, conceptual understanding) undergo long periods of stasis interrupted by rapid periods of change. The rediscovery of Mendel’s laws led to a conceptual shift during the beginning of the 20th century. Likewise, the discovery of restriction enzymes more than 30 years ago prompted another conceptual shift. These enzymes are just tools, yet they helped shift biology from a largely descriptive science to a manipulative one. Today, bioinformatics is helping to propel biology toward yet another conceptual shift. “What it does is to provide a very finely honed set of tools for people to turn biological questions into molecular terms,” says John Sulston, director of the Sanger Centre (Ferry 1998). Biologists are familiar with the terms “*in vivo*” and “*in vitro*,” used to describe processes that occur in the body and in the test tube. Now, they are becoming acquainted with a third term, “*in silico*,” used to describe a new branch of biology that requires little more than a computer and a connection to the Internet.

As more and more DNA and protein sequence data find their way into computer databases, the ability of bioinformatics to address biological questions becomes ever more powerful. The amount of genetic data available and the rate of its acquisition is by any measure astonishing. According to Collins, it took four years to obtain the first one billion base-pairs of human sequence and just four months to get the second billion (Marshall et al. 2000). As of

May 2001, there have been 51 microbial genomes completely sequenced, 16 more are being annotated, and an additional 86 are in the process of being sequenced. In addition, nine eukaryotic genomes are largely finished, including: yeast, the nematode worm *C. elegans*, the fruit fly *Drosophila melanogaster*, mouse, and human.

Bioinformatics is leading a shift away from the reductionism that characterized molecular biology of the last century and toward a more holistic approach. Technologies such as DNA microarrays permit scientists to glimpse the dynamic nature of genomes by examining the expression of thousands of genes at a time, as opposed to the traditional piecemeal approach.

These genetic data represent a treasure trove for researchers and companies interested in a wide variety of applications. Genomic scientists are often described as engaging in genetic prospecting or in gene mining. There is ample justification to regard these data as the genetic motherlode. Approximately one-third of the genes identified in the newly sequenced genomes are of unknown function and therefore of academic and potentially economic interest. New companies are formed on a monthly basis to take part in this genetics sweepstakes. The oldest of the bioinformatics companies are only about five years old. Currently, there are 14 such companies being traded publicly. They have a combined market capitalization of \$10 billion and are growing at a rapid pace (Pickering 1999).

Genetics’ Swiss Army Knife

If bioinformatics is considered to be a tool, then it can be described as a Swiss Army knife, having blades with many different uses.

Basic Sciences

To date, gene maps are available for a wide variety of organisms including human, mouse, and about 30 other mammalian species. These maps are used to identify genes for heritable traits and to unravel patterns of genome organization. Comparisons of genomic sequences between different organisms can help clarify evolutionary relationships. Such molecular data caused a stir in 1990 when they suggested that humans and chimpanzees are more closely related to each other than either is to gorillas (Goodman et al. 1990). More recently, DNA sequence analysis suggested that Neanderthals do not represent a

link in human evolution (Krings et al. 1997; Ovchinnikov et al. 2000)

Forensic Science

Minisatellites consist of short DNA sequences that repeat in tandem. The number of repeats and the sequence within each repeat can exhibit wide variation in a population. This aspect of genome organization was exploited by Alec Jeffreys to create "DNA fingerprinting." His technique has revolutionized identity testing. To maximize the impact of DNA testing in forensics, the FBI, together with individual states, established the Combined DNA Index System (CODIS). Currently, this genetic database contains profiles from more than 230,000 convicted offenders and identifies a perpetrator for about every 1,000 offender samples in the system (Asplen 1999). DNA testing is now the standard technique for confirming paternity. Its use, however, is not restricted to establishing paternity in the present. As long as suitable DNA samples are available, DNA tests can challenge our view of history. For example, recent DNA tests suggest that Thomas Jefferson fathered at least one of his slave Sally Heming's six children (Foster et al. 1998).

Agriculture

Bioinformatics is playing an increasing role in agriculture. Genome projects for major crop plants such as rice and corn are well underway, and these data are being directed toward crop improvement and pest control. Some current gene discovery efforts are aimed at improving seed quality, engineering plants to cope with metals, and manipulating plant micronutrients for human health – an application called "nutritional genomics" (DellaPenne 1999). There are also genome projects for a number of agriculturally important animals including cattle, pig, horse, sheep and chicken. These genetic data are being sifted to locate genes associated with hereditary disorders, infectious disease resistance, and desirable phenotypes called ETLs (economic trait loci) (Womack & Kata 1995).

Medicine

The ability to correlate genetic data with medical records promises to improve our understanding of disease and improve treatments. Already, DNA arrays are being used to classify cancers by monitoring their patterns of gene expression (Golub et al. 1999). An important resource will be a data-

base of single nucleotide polymorphisms (SNPs). The goal is to collect perhaps 100,000 to 300,000 single-base variations from several hundred people representing diverse ethnic backgrounds. These data will serve as landmarks for a high resolution genome map that will help scientists identify genes which cannot be detected through family studies. SNP data are expected to increase our understanding of human diversity and usher in an era of personalized medicine. The economic potential of such databases is so attractive that at least three private companies, a nonprofit consortium of pharmaceutical companies and Britain's Wellcome Trust, and the National Institutes of Health each are creating their own SNP databases (Roberts 2000).

Pharma/Biotechnology

Pharmacogenetics focuses on genetic polymorphisms that produce variation in patients' responses to medications. Currently, there are over 500,000 human DNA sequences contained in genetic databases. It is estimated that these may give rise to 150,000 targets for drug development (Liszewski 2000). The tools of functional genomics are being used to establish the metabolic roles played by these candidate gene products. Already, some scientists are calling for an initiative in structural biology, somewhat reminiscent of the Human Genome Project, that would produce the three-dimensional structures for the estimated 1,000 to 5,000 distinct spatial arrangements taken on by polypeptides found in nature.

Environmental Science

Scientists concerned with loss of biodiversity are using bioinformatics to establish the extent of genetic variation within populations. Within captive breeding programs, these data are used to determine which matings will increase the population's diversity. Microbial genomes also are proving valuable to environmental efforts, such as bioremediation. The genome of *Deinococcus radiodurans*, the most radiation-resistant bacterium known, has been completely sequenced and is being tested for use in the clean up of contaminated radiation spills (White et al. 1999). Genomic data from bacteria are helping scientists assess the biological contributions to ancient and modern geochemical cycles. Bioinformatics helps provide a deeper understanding of such environmentally important processes as iron and sulfur oxidation, nitrogen

fixation, carbon fixation, and metal and acid resistance (Banfield & Marshall 2000).

Constraints & Ethical Concerns

The rapid development of bioinformatics is not happening without growing pains, however. Perhaps the biggest problem facing this emerging discipline is the lack of biologists with computing expertise. In nearly all areas of biology, there is a glut of Ph.D.s. A study carried out by the National Research Council in 1998 confirmed the suspicion of many young scientists that the number of newly minted Ph.D.s greatly exceeds the supply of desirable jobs. "I call it the La Guardia effect," said Shirley Tilghman, a molecular biologist at Princeton University who chaired the NRC panel. She sees "a lot of trained scientists who are circling, burning up very important and useful fuel, and waiting for their turn to land" (Holden 1998).

The situation in bioinformatics is very different. "You can count on the fingers of one hand" the number of researchers with topflight training in both fields, according to geneticist David Botstein of Stanford University who co-chaired a recent advisory panel that looked into the status of biocomputing in the United States (Malakoff 1999). This 16-member working group was appointed by Harold Varmus in 1998 to help NIH develop a strategy for dealing with the lack of expertise in bioinformatics. The panel recommended that NIH create between five and 20 training and research centers in biocomputing. These centers would be located at universities and independent research institutes around the country and would help reduce the expertise bottleneck currently constraining the growth of bioinformatics.

Although bioinformatics is intimately associated with genetic research, it does not evoke the sort of ethical concern that accompanies applications such as cloning or genetically modified food. Recent surveys show that people from industrial countries value the use of genetic technology in medical research over any other type of application (Richmond et al. 1999). Nevertheless, the establishment of genetic databases is not without its attendant controversy. For the past two years, Iceland's small scientific community has been embroiled in a heated debate regarding plans to pool the entire country's medical records into a database that will be used by a private company to identify disease-causing genes. Of particular concern to many is the method by which informed consent is obtained – or not obtained, as its critics contend. Unlike the usual

procedure where each individual signs a consent form, citizens of Iceland automatically have their records included in the database, unless they specifically ask to be left out. The general population seems to be less troubled by the controversy than the scientific community. Thus far, only 5 percent of the people have asked to have records excluded from the database (Enserink 2000).

Iceland is not alone in its plans to establish a national gene bank. A group of geneticists in Estonia recently presented a plan to its government seeking to establish a genetic database that contains genotypes of at least 70 percent of the country's 1.4 million people (Frank 1999). Critics of the proposed Estonian database complain that expensive high-tech genetic research will not benefit their population as much as focusing on simple lifestyle factors such as diet, smoking, and alcohol abuse. Sweden too is facing criticisms. This time over its public tissue banks. These collections of blood, sperm, fertilized eggs, and biopsies were originally saved to help diagnose and treat the patients themselves. Now, biotech companies want access to the samples and the patients' medical records. In response to public concern over reported breaches of privacy, the Swedish Medical Research Council has established regulations that require users of banked samples to obtain informed consent for each new use of the tissue samples (Nilsson & Rose 1999).

Americans are more accepting of genetic technology than Europeans, however they share similar concerns. A nationwide survey commissioned by the National Center for Genome Resources was completed in 1998. Results showed that 69 percent of respondents thought that health and insurance companies should not have access to genetic information, 85 percent thought that employers should not have such access, and 65 percent would withhold such information from relatives (Richmond et al. 1999). Even databases containing anonymized samples have ethical concerns. Aside from the worry that hackers might reconstruct someone's identity, such studies may produce information that is of potential benefit to the individual (such as predisposition to a disease for which preventative measures are available).

Privacy issues also surround the Combined DNA Index System (CODIS) established by the FBI. CODIS is used as an investigative tool, comparing DNA sequences found at crime scenes with those of convicted offenders in the database. Former Attorney General Janet Reno identified several privacy concerns that pertain to the construction and

use of this database. Currently, DNA samples are collected, tested, and then stored indefinitely. The concern is that such samples will be tested in the future for purposes other than identification. Another worry involves testing of samples obtained from persons arrested but not convicted of a crime. Does such testing violate their civil liberties?

Finally, DNA testing can be used with samples that are many years old. The Innocence Project, based at Cardozo School of Law in New York City, has used DNA testing to free 67 men wrongly convicted of crimes. The average age of the DNA samples tested was 9.56 years. Currently just two states, New York and Illinois, have laws that authorize postconviction DNA testing for inmates (Sheck et al. 2000). Does this mean we should reexamine the statute of limitations? Questions such as these have no simple answers, and it likely will take years for society to reach a consensus.

Bioinformatics in the Classroom

Bioinformatics is a young science, still undergoing development. It contains, however, a well-accepted body of knowledge about genomes that justifies treatment in the high school biology curriculum. In general, there is a lag, measured in years, between scientific advance and its inclusion in the curriculum. For example, the polymerase chain reaction (PCR) was introduced to the scientific community in 1985 (Saiki et al. 1985). Biology teachers became aware of the technique through stories in the media and wanted to learn more about it. It was not until 1990, however, when PCR inventor Kary Mullis published an article about the technique in *Scientific American* that teachers found a treatment of this important topic that was accessible to them (Mullis 1990). It took another couple of years for PCR to be mentioned in most high school biology textbooks.

Bioinformatics is currently in this lag phase. Despite the importance of the Human Genome Project, current high school biology textbooks (which are 600 to 1,000 pages long) devote just a

page or two to the topic and limit their discussion almost exclusively to the quest to identify disease genes. If discussed at all, the impact of genomics on the study of cell biology, evolution, and applications to areas other than medicine is given superficial treatment. Likewise, the use of computers to acquire, store and manipulate genetic data rates barely a mention.

Computers in the classroom are receiving increasing support, and the education literature reflects this trend. Articles explaining how to integrate computer use in teaching are becoming commonplace. Typically, such articles take the form of primers intended to educate teachers about how to use the World Wide Web and to suggest which sites to visit (Francis 2000). The use of computers as an investigative tool is seldom discussed. When articles about bioinformatics are directed to teachers, they usually are aimed at the undergraduate level (Seitz & Leake 1999). Often, these papers are intimidating to teachers. They frequently use jargon (such as ESTs, SSTs, BLAST) with little or no explanation and attempt to have the reader jump in and use databases created by and for the bioinformatics community.

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school biology teachers is found at the Web site of the DNA Learning Center (DNALC) at Cold Spring Harbor Laboratory (<http://vector.cshl.org>). Among its resources are *DNA from the Beginning*, a genetics primer, and the *Biology Animation Library* which allows teachers to play or download animations which depict DNA-related concepts or techniques.

The DNALC also has developed a relatively simple system to allow students to obtain and work with their own DNA sequences as an engaging entry into bioinformatics. Using a kit developed by the DNALC and distributed by Carolina Biological Supply Company, students isolate their DNA from hair roots or cheek cells. The DNA is mixed with freeze-dried reagents and taken through the polymerase chain reaction to amplify (enrich) a small region of their genome. Amplified DNA samples are sent to the DNALC where the DNA sequences are obtained and posted at the *Sequence Server* facility at the DNALC web site (<http://vector.cshl.org/sequences/>). The *Sequence Server* features an easy-to-use interface that allows students to launch database searches, sequence alignments, and phylogenetic analyses from a centralized work space. These tools enable students to use their own DNA data to explore human population genetics and test theories of human evolution – including comparisons with DNA from humans and Neanderthals. A student also may use her/his DNA sequence to perform a search to find similar sequences in the GenBank sequence database. Sequences so identified then can be moved to the student's workspace for further analysis.

The DNALC web site also features a resource called *Genexercises* (<http://vector.cshl.org/genexercises/>). This bioinformatics tool provides students an interactive “walk” through a forensic DNA case study and an analysis of human molecular evolution. Each case includes a contextual narrative, interactive questions, and choices of sequences to be submitted to the *Sequence Server* for analysis. The evolution case compares human, chimpanzee and Neanderthal sequences to determine whether Neanderthal was the direct ancestor of modern humans. The forensic case identifies the remains of the Romanov royal family (Nicholas Romanov was the last tsar of Russia) and tests the claim of a woman named Anna Anderson who believed she was the Duchess Anastasia.

Conclusion

Biology teachers can use bioinformatics to relate concepts from biology, mathematics, and computer

science and to articulate them with the ethical and social concerns raised by genetic technology. If society is to gain the most from genomic biology, then the public must be able to rationally consider scientific issues. They should neither place a blind trust in scientists, nor should they dismiss new technologies out of hand. This point also is made in the *National Science Education Standards*:

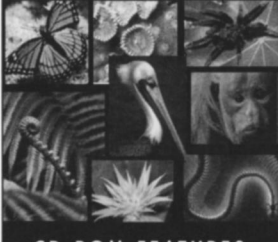
“Because molecular biology will continue into the 21st century as a major frontier of science, students should understand the chemical basis of life, not only for its own sake, but because of the need to take informed positions on some of the practical and ethical implications of humankind's capacity to tinker with the fundamental nature of life” (NRC 1996).

The days when scientists could clone a gene and publish a journal article about it are nearing an end. The era of bioinformatics is characterized by whole genome sequencing, DNA microarrays, and *in silico* biology. Welcome to the revolution!

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